MONOMORPHISMS INTO COPRODUCTS OF SPECTRA OF FIELDS

THE STACKS PROJECT AUTHORS

Editorial note. Counted span is the author's own proof environment, source lines 4911-4922, following the statement at 4897-4909.

Editorial note. The core algebra step is concluded by the named cross-chapter engine algebra-lemma-epimorphism-over-field (tag 04VV, present in the local tag index); the 'This holds for dimension reasons' phrase points at that engine.

Editorial note. The final subset assembly is mechanical from the per-component dichotomy.

Allowed standard prerequisites (editorial).

- Ring epimorphisms over a field. Algebra, Lemma algebra-lemma-epimorphism-over-field (cross-chapter, tag 04VV, cited at ~4921). Engine: k -> R with R tensor_k R = R forces R = k or R = 0.

Source reference. Original reference 04VV: focused-seeds/geometry/S9-schemes-mono-towards-spec-field/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let $k_1, \ldots, k_n$ be fields. For any monomorphism of schemes $X \to \mathrm{Spec}(k_1 \times \ldots \times k_n)$ there exists a subset $I \subset \{1, \ldots, n\}$ such that $X \cong \mathrm{Spec}(\prod_{i \in I} k_i)$ as schemes over $\mathrm{Spec}(k_1 \times \ldots \times k_n)$. More generally, if $X = \coprod_{i \in I} \mathrm{Spec}(k_i)$ is a disjoint union of spectra of fields and $Y \to X$ is a monomorphism, then there exists a subset $J \subset I$ such that $Y = \coprod_{i \in J} \mathrm{Spec}(k_i)$.*

Proof. First reduce to the case $n = 1$ (or $\#I = 1$) by taking the inverse images of the open and closed subschemes $\mathrm{Spec}(k_i)$. In this case X has only one point hence is affine. The corresponding algebra problem is this: If $k \to R$ is an algebra map with $R \otimes_k R \cong R$, then $R \cong k$ or $R = 0$. This holds for dimension reasons. See also Algebra, Lemma 04VV $\square$

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